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United States Patent	12410025
Kind Code	B2
Date of Patent	September 09, 2025
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Sheet storage device and image formation system including same

Abstract

A sheet storage device includes a regulation cursor and a hold mechanism. The hold mechanism includes a guide shaft, and a stopper having a guide hole in which the guide shaft is inserted, and being capable of reciprocating with the regulation cursor. When the stopper is inclined to one side, movement of the regulation cursor is restricted. When the stopper is raised to the other side, movement of the regulation cursor is enabled. The stopper has a contact part that contacts the regulation cursor, so as to restrict the stopper from inclining to the other side, when the regulation cursor is enabled to move.

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Appl. No.:	18/731594
Filed:	June 03, 2024

Prior Publication Data

Document Identifier	Publication Date
US 20240400327 A1	Dec. 05, 2024

Foreign Application Priority Data

JP	2023-092369	Jun. 05, 2023
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Publication Classification

Int. Cl.: B65H1/26 (20060101); B65H1/04 (20060101); B65H1/14 (20060101)

U.S. Cl.:

CPC **B65H1/266** (20130101); **B65H1/04** (20130101); B65H1/14 (20130101); B65H2220/01 (20130101); B65H2220/04 (20130101); B65H2403/411 (20130101); B65H2405/1142 (20130101); B65H2511/12 (20130101); B65H2511/20 (20130101); B65H2511/22 (20130101); B65H2701/1315 (20130101)

Field of Classification Search

CPC: B65H (9/101); B65H (2403/411); B65H (2511/12); B65H (2511/11); B65H (2405/1122); B65H (2405/1142); B65H (2405/11425); B65H (1/266)

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Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
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Patent No.	Application Date	Country	CPC
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Background/Summary

INCORPORATION BY REFERENCE

(1) This application is based upon and claims the benefit of priority from the corresponding Japanese Patent Application No. 2023-092369 filed Jun. 5, 2023, the entire contents of which are hereby incorporated by reference.

BACKGROUND

(2) The present disclosure relates to a sheet storage device and an image formation system including the same.

(3) A conventional sheet storage device is equipped with a regulation cursor that regulates positional deviation of a sheet. The regulation cursor can be moved by a user's operation, and stops in a state where it contacts an edge of the sheet, so as to regulate positional deviation of the sheet.

(4) A conventional regulation cursor is capable of engaging with a concave-convex shape formed at a predetermined pitch in a moving direction of the regulation cursor, for example. By allowing the regulation cursor to engage with the concave-convex shape, the regulation cursor can be held in a stationary state.

SUMMARY

(5) A sheet storage device according to a first aspect of the present disclosure includes a setting plate, a regulation cursor, and a hold mechanism. The setting plate is used for setting sheets thereon. The regulation cursor is supported in a manner capable of reciprocating in a horizontal direction, and contacts an edge of the sheet on the setting plate from a predetermined direction, so as to regulate positional deviation of the sheet in the predetermined direction. The hold mechanism

stationarily holds the regulation cursor. The hold mechanism includes a guide shaft and a stopper. The guide shaft extends in the predetermined direction. The stopper is linked with the regulation cursor, and has a guide hole in which the guide shaft is inserted, so as to be capable of reciprocating with the regulation cursor along the guide shaft. The stopper is capable of inclining to one side in the predetermined direction on a fulcrum of a hole side end part having the guide hole. When the stopper is inclined to the one side, an inner edge of the guide hole contacts the guide shaft so as to generate frictional resistance, which restricts movement of the regulation cursor with the stopper, and when the stopper is raised to the other side opposite to the one side from the state where movement of the regulation cursor is restricted, the regulation cursor is enabled to move with the stopper. The stopper has a contact part. The contact part contacts the regulation cursor, so as to restrict the stopper from inclining to the other side, when the regulation cursor is enabled to move.

(6) An image formation system according to a second aspect of the present disclosure includes the sheet storage device described above, and an image forming apparatus. The image forming apparatus is linked with the sheet storage device. The sheet storage device feeds the image forming apparatus with the sheet. The image forming apparatus performs printing on the sheet fed from the sheet storage device.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a perspective view of a sheet storage device according to an embodiment.
- (2) FIG. 2 is a schematic diagram of an image formation system including the sheet storage device according to the embodiment.
- (3) FIG. 3 is a perspective view of the sheet storage device according to the embodiment, without an outer cover.
- (4) FIG. 4 is a perspective view of an elevating mechanism of a setting plate in the sheet storage device according to the embodiment.
- (5) FIG. 5 is a perspective view of a winding motor and its vicinity of the elevating mechanism illustrated in FIG. 4.
- (6) FIG. 6 is a plan view of a sheet storage region of the sheet storage device according to the embodiment, viewed from above.
- (7) FIG. 7 is a schematic diagram (a plan view viewed from above) of a mechanism for moving a pair of width direction regulation cursors of the sheet storage device according to the embodiment.
- (8) FIG. 8 is a perspective view of the width direction regulation cursors and their vicinity of the sheet storage device according to the embodiment, viewed from front.
- (9) FIG. 9 is a perspective view of a stopper of the sheet storage device according to the embodiment.
- (10) FIG. 10 is a perspective view of a state where the stopper illustrated in FIG. 9 is linked with the width direction regulation cursor.
- (11) FIG. 11 is a cross-sectional view schematically illustrating a state where the stopper of the sheet storage device according to the embodiment is inclined.
- (12) FIG. 12 is a cross-sectional view schematically illustrating a state where the stopper of the sheet storage device according to the embodiment is raised.

DETAILED DESCRIPTION

- (13) Hereinafter, with reference to FIGS. 1 to 12, a sheet storage device **100** according to an embodiment of the present disclosure and an image formation system **100S** including the same are described.
- (14) <Structure of Image Formation System>
- (15) The sheet storage device **100** has an appearance as illustrated in FIG. 1. The sheet storage

device **100** is linked with an image forming apparatus **1000** as illustrated in FIG. 2. When being linked with the image forming apparatus **1000**, the sheet storage device **100** constitutes the image formation system **100S** with the image forming apparatus **1000**. FIG. 2 is a schematic diagram of the image formation system **100S** (i.e., the sheet storage device **100** linked with the image forming apparatus **1000**). FIG. 2 corresponds to a case where the image formation system **100S** is viewed from front.

(16) The sheet storage device **100** is installed with the image forming apparatus **1000** on a floor. The direction perpendicular to the floor (a flat surface) on which the sheet storage device **100** is installed corresponds to an “up and down direction”.

(17) Here, in horizontal directions perpendicular to the up and down direction, one direction corresponds to a “first direction”, and the other direction perpendicular to the one direction corresponds to a “second direction”. In the following description, the one direction of the horizontal directions is denoted by **D1** and is simply referred to as a first direction **D1**, while the other direction of the horizontal directions is denoted by **D2** and is simply referred to as a second direction **D2**.

(18) The first direction **D1** corresponds to a front and back direction when viewing the sheet storage device **100** from front. One side in the first direction **D1** is a front side (i.e., a near side when viewing the device from front), and the other side opposite to the one side in the first direction **D1** is a back side (i.e., a far side when viewing the device from front).

(19) The second direction **D2** corresponds to a left and right direction when viewing the sheet storage device **100** from front. One side in the second direction **D2** is a right side, and the other side opposite to the one side in the second direction **D2** is a left side.

(20) Note that the first direction **D1** and the second direction **D2** each correspond to a “predetermined direction”. The front side in the first direction **D1** corresponds to “one side in a predetermined direction”, and the back side in the first direction **D1** corresponds to “the other side in a predetermined direction”. The right side in the second direction **D2** corresponds to “one side in a predetermined direction”, and the left side in the second direction **D2** corresponds to “the other side in a predetermined direction”.

(21) The sheet storage device **100** is attachable and detachable from the image forming apparatus **1000**. The sheet storage device **100** may be an optional device. The sheet storage device **100** is disposed on the right side of the image forming apparatus **1000**.

(22) The sheet storage device **100** stores sheets **S**. For instance, the sheet **S** is a paper sheet. The sheet storage device **100** feeds the sheet **S** to the image forming apparatus **1000**. In other words, the sheet storage device **100** functions as a feeding device.

(23) On the right side of the image forming apparatus **1000**, the sheet storage device **100** feeds the sheet **S** to the image forming apparatus **1000**, from the right side to the left side. In other words, in the second direction **D2**, the direction from the right side to the left side is a feeding direction of the sheet **S** from the sheet storage device **100** to the image forming apparatus **1000**. In this structure, the first direction **D1** is a direction perpendicular to the feeding direction of the sheet **S**.

(24) The image forming apparatus **1000** prints an image on the sheet **S** fed from the sheet storage device **100**. A printing method of the image forming apparatus **1000** is an electrophotographic method. It may also be possible that the printing method of the image forming apparatus **1000** is an inkjet method.

(25) <Structure of Image Forming Apparatus>

(26) The image forming apparatus **1000** conveys the sheet **S** fed from the sheet storage device **100**, along a conveying path. The image forming apparatus **1000** prints an image on the sheet **S** under conveyance. Note that the conveying path of the sheet **S** is shown by a broken line with an arrow in FIG. 2.

(27) The image forming apparatus **1000** includes a photosensitive drum **1001** and a transfer roller **1002**. The photosensitive drum **1001** carries a toner image on its outer circumference surface. The

transfer roller **1002** is pressed to contact the photosensitive drum **1001**, so as to form a transfer nip with the photosensitive drum **1001**. The transfer roller **1002** rotates together with the photosensitive drum **1001**. When the sheet S under conveyance enters in the transfer nip, a transfer process is performed on the sheet S, and the toner image is transferred onto the sheet S.

(28) Although not illustrated, the image forming apparatus **1000** further includes a charging device, an exposure device, and a development device. The charging device charges the outer circumference surface of the photosensitive drum **1001**. The exposure device forms an electrostatic latent image on the outer circumference surface of the photosensitive drum **1001**. The development device develops the electrostatic latent image on the outer circumference surface of the photosensitive drum **1001** into the toner image.

(29) The image forming apparatus **1000** includes a fixing roller pair **1003**. The fixing roller pair **1003** includes a heating roller and a pressure roller. A heater is embedded in the heating roller. The pressure roller is pressed to contact the heating roller so as to form a fixing nip with the heating roller. The pressure roller rotates with the heating roller. When the sheet S under conveyance enters the fixing nip, a fixing process is performed on the sheet S, and the toner image is fixed to the sheet S. The sheet S after passing through the fixing nip is discharged onto a discharge tray ET.

(30) Note that the image forming apparatus **1000** includes a feed roller **1004**. The feed roller **1004** feeds the sheet S to the conveying path from a sheet cassette CA disposed in a main body of the image forming apparatus **1000**. In other words, the image forming apparatus **1000** can also print on the sheet S set in the sheet cassette CA.

(31) In addition, the image forming apparatus **1000** includes an image reading device **1005**. The image reading device **1005** reads an original so as to generate image data of the original. The image forming apparatus **1000** can print an image based on the image data of the original on the sheet S (i.e., can copy the original).

(32) <Structure of Sheet Storage Device>

(33) As illustrated in FIG. 3, the sheet storage device **100** has a front frame Ff and a rear frame Fr. The front frame Ff and the rear frame Fr are made of sheet metal. The front frame Ff is disposed on the front side, and the rear frame Fr is disposed on the back side. The front frame Ff has a front surface part whose thickness direction is the first direction D1, and the rear frame Fr has a rear surface part whose thickness direction is the first direction D1. The front frame Ff and the rear frame Fr are disposed in such a manner that the front surface part and the rear surface part face each other in the first direction D1.

(34) The sheet storage device **100** has a sheet storage region, which is a region between the front frame Ff and the rear frame Fr in the first direction D1. The sheet storage region is covered with a top cover CV from above (see FIGS. 1 and 2).

(35) The top cover CV is supported in a pivotable manner with respect to the front frame Ff and the rear frame Fr. The top cover CV is pivotable about an axis extending in the first direction D1. The top cover CV pivots on a fulcrum of the left side end part so as to move the right side end part upward and downward. In other words, the top cover CV opens and closes an upper opening of the sheet storage region.

(36) When the right side end part of the top cover CV is moved upward (in other words, when opening the top cover CV), the upper opening of the sheet storage region is exposed. Further, in the state where the top cover CV is opened, the sheets S are put in the sheet storage region through the upper opening of the sheet storage region. In this state, the right side end part of the top cover CV is moved downward (in other words, the top cover CV is closed), and thus the upper opening of the sheet storage region is closed.

(37) The sheet storage device **100** includes a setting plate **1**. The setting plate **1** is made of metal sheet. The setting plate **1** is disposed in the sheet storage region. The sheets S are set on the setting plate **1**. The sheets S are stacked in the up and down direction. The setting plate **1** is supported in a manner capable of reciprocating in the up and down direction.

(38) The sheet storage device **100** includes an elevating mechanism **2**. A structure of the elevating mechanism **2** is illustrated in FIGS. **4** and **5**. The elevating mechanism **2** moves the setting plate **1** in the up and down direction. The elevating mechanism **2** moves the setting plate **1** upward, so as to keep a position in the up and down direction of the top one of the sheets **S** set on the setting plate **1**.

(39) The elevating mechanism **2** includes wires **20** linked with the setting plate **1**. The wires **20** include a first wire **201** whose one end is linked with a right front side end part of the setting plate **1**, a second wire **202** whose one end is linked with a left front side end part of the setting plate **1**, a third wire **203** whose one end is linked with a right back side end part of the setting plate **1**, and a fourth wire **204** whose one end is linked with a left back side end part of the setting plate **1**. The elevating mechanism **2** winds or unwinds the wires **20** so as to move the setting plate **1** upward or downward.

(40) The elevating mechanism **2** includes winding drums **21**, relay pulleys **22**, and a winding motor **M**. The winding drums **21** includes a first winding drum **211** linked with the other end of the first wire **201**, a second winding drum **212** linked with the other end of the second wire **202**, a third winding drum **213** linked with the other end of the third wire **203**, and a fourth winding drum **214** linked with the other end of the fourth wire **204**. When each of the first to fourth winding drums **211** to **214** rotates forward, it winds the wire **20** linked with itself, while when it rotates backward, it unwinds the wire **20** linked with itself.

(41) The first winding drum **211** and the second winding drum **212** are disposed outward in the first direction **D1** of the front frame **Ff**. The third winding drum **213** and the fourth winding drum **214** are disposed outward in the first direction **D1** of the rear frame **Fr**. Note that “outward in the first direction **D1**” means the direction from the center of sheet storage region in the first direction **D1** toward the front side or the back side in the first direction **D1**. In other words, “outward in the first direction **D1** of the front frame **Ff**” means the opposite side of the sheet storage region side in the first direction **D1** of the front frame **Ff** (i.e., the front side of the front frame **Ff**), and “outward in the first direction **D1** of the rear frame **Fr**” means the opposite side of the sheet storage region side in the first direction **D1** of the rear frame **Fr** (i.e., the back side of the rear frame **Fr**).

(42) The first winding drum **211** and the second winding drum **212** are arranged in this order from the front frame **Ff** side outward in the first direction **D1**. The fourth winding drum **214** and the third winding drum **213** are arranged in this order from the rear frame **Fr** side outward in the first direction **D1**.

(43) The first to fourth winding drums **211** to **214** are supported in a rotatable manner about the same axis extending in the first direction **D1**. Specifically, the first to fourth winding drums **211** to **214** are attached to the same rotation shaft **210**, and rotate together with the rotation shaft **210**. One end part of the rotation shaft **210** in the axis direction is supported by the front frame **Ff** in a rotatable manner, and the other end part in the axis direction opposite to the one end part is supported by the rear frame **Fr** in a rotatable manner, so that the rotation shaft **210** extends in parallel to the first direction **D1**. The first winding drum **211** and the second winding drum **212** are disposed in the right side end part of the front frame **Ff** in the second direction **D2**, while the third winding drum **213** and the fourth winding drum **214** are disposed in the right side end part of the rear frame **Fr** in the second direction **D2**. In other words, one end part of the rotation shaft **210** is supported in a rotatable manner by the right side end part of the front frame **Ff** in the second direction **D2**, and the other end part of the rotation shaft **210** is supported in a rotatable manner by the right side end part of the rear frame **Fr** in the second direction **D2**.

(44) When the winding motor **M** is driven, it rotates the rotation shaft **210**. When the rotation shaft **210** rotates, the first to fourth winding drums **211** to **214** rotate. The winding motor **M** is disposed outward in the first direction **D1** of the rear frame **Fr**. The winding motor **M** is linked with the back side end part of the rotation shaft **210** with a transmission mechanism **23** including a plurality of gears such as a screw gear.

(45) Three relay pulleys **22** are disposed outward in the first direction **D1** of the front frame **Ff**, and

three relay pulleys **22** are disposed outward in the first direction of the rear frame **Fr**. The three relay pulleys **22** on the front side are supported by the front frame **Ff** in a rotatable manner, and the three relay pulleys **22** on the back side are supported by the rear frame **Fr** in a rotatable manner. Each of the relay pulleys **22** is disposed higher than the rotation shaft **210** in the up and down direction.

(46) The first wire **201** extends from the setting plate **1**, wraps around the one relay pulley **22**, and reaches the first winding drum **211**. The second wire **202** wraps around the two relay pulleys **22**. One relay pulley **22** for the second wire **202** is positioned at the left side of the other relay pulley **22** for the second wire **202** in the second direction **D2**. The other relay pulley **22** for the second wire **202** rotates about the same axis as the relay pulley **22** for the first wire **201**. The second wire **202** extends from the setting plate **1**, wraps around the two relay pulleys **22**, and reaches the second winding drum **212**.

(47) The third wire **203** extends from the setting plate **1**, wraps around the one relay pulley **22**, and reaches the third winding drum **213**. The fourth wire **204** wraps around the two relay pulleys **22**. One relay pulley **22** for the fourth wire **204** is positioned at the left side of the other relay pulley **22** for the fourth wire **204** in the second direction **D2**. The other relay pulley **22** for the fourth wire **204** rotates about the same axis as the relay pulley **22** for the third wire **203**. The fourth wire **204** extends from the setting plate **1**, wraps around the two relay pulleys **22**, and reaches the fourth winding drum **214**.

(48) When the winding motor **M** rotates forward, the winding drum **21** winds the wire **20**. In this way, the setting plate **1** moves upward. On the other hand, when the winding motor **M** rotates backward, the wire **20** is unwound from the winding drum **21**. In this way, the setting plate **1** moves downward.

(49) As illustrated in FIGS. **2** and **3**, the sheet storage device **100** includes a feeding unit **3**. The feeding unit **3** feeds the sheet **S** from the sheet storage device **100** (i.e., the sheet storage region) to the image forming apparatus **1000**. The feeding unit **3** includes a pickup roller **31** and a conveying roller pair **32**.

(50) The pickup roller **31** is supported in a rotatable manner. The pickup roller **31** is disposed at a position capable of contacting a downstream side end part of the sheet **S** in the feeding direction from above, in the state where the sheet **S** is set on the setting plate **1**. The pickup roller **31** is supported in a rotatable manner. The pickup roller **31** contacts the sheet **S** set on the setting plate **1** from above and rotates in this state, so as to pull out the sheet **S** from the setting plate **1**.

(51) The conveying roller pair **32** is supported in a rotatable manner. The conveying roller pair **32** is a pair of rollers that are pressed to contact each other. The conveying roller pair **32** is disposed on the downstream side of the pickup roller **31** in the feeding direction. The conveying roller pair **32** pinches the sheet **S** pulled out from the setting plate **1** by the pickup roller **31**, and rotates. In this way, the conveying roller pair **32** conveys the sheet **S** to the image forming apparatus **1000**. The number of disposed conveying roller pairs **32** is not particularly limited, but it can be changed depending on factors such as a conveying path length for the sheet **S** from the pickup roller **31** to the image forming apparatus **1000**.

(52) The sheet **S** on the setting plate **1** is sent out from the right side to the left side in the second direction **D2**. In other words, on the setting plate **1**, the left side end of the sheet **S** in the second direction **D2** is a front end, while the right side end of the sheet **S** in the second direction **D2** is a rear end. In addition, on the setting plate **1**, the first direction **D1** (i.e., the front and back direction) is a width direction of the sheet **S**.

(53) Note that the elevating mechanism **2** moves the setting plate **1** upward, so as to allow the pickup roller **31** to contact the top one of the sheets **S** on the setting plate **1**. In a case where the sheets **S** are fed continuously from the sheet storage device **100** to the image forming apparatus **1000**, the elevating mechanism **2** repeats moving the setting plate **1** upward and stopping the same. In this way, the contact between the pickup roller **31** and the top one of the sheets **S** on the setting

plate **1** is maintained.

(54) As illustrated in FIG. **6**, the sheet storage device **100** includes width direction regulation cursors **4A** and **4B**, and a rear end regulation cursor **4C**. The width direction regulation cursor **4A**, the width direction regulation cursor **4B**, and the rear end regulation cursor **4C** are supported in a manner capable of reciprocating in predetermined directions perpendicular to the up and down direction (i.e., in the horizontal direction). The width direction regulation cursor **4A**, the width direction regulation cursor **4B**, and the rear end regulation cursor **4C** are moved in predetermined directions by user's operation. Then, the width direction regulation cursor **4A**, the width direction regulation cursor **4B**, and the rear end regulation cursor **4C** contact edges of the sheet **S** set on the setting plate **1**, from predetermined directions. In this way, the width direction regulation cursor **4A**, the width direction regulation cursor **4B**, and the rear end regulation cursor **4C** regulate positional deviations in predetermined directions of the sheet **S** on the setting plate **1**.

(55) The width direction regulation cursors **4A** and **4B** move in the first direction **D1**, and contact the edges of the sheet **S** on the setting plate **1** in the first direction **D1**, so as to regulate a positional deviation in the first direction **D1** of the sheet **S** on the setting plate **1**. Note that the width direction regulation cursors **4A** and **4B** sandwich the sheet **S** on the setting plate **1** in the first direction **D1**. The rear end regulation cursor **4C** moves in the second direction **D2**, and contacts the edge of the sheet **S** on the setting plate **1** in the second direction **D2**, so as to regulate a positional deviation in the second direction **D2** of the sheet **S** on the setting plate **1**.

(56) Note that when viewing from above, the setting plate **1** has openings (no numeral) on moving paths of the width direction regulation cursors **4A** and **4B**, and an opening (no numeral) on a moving path of the rear end regulation cursor **4C**. Therefore, when the width direction regulation cursors **4A** and **4B** move in the first direction **D1**, they do not contact the setting plate **1**. When the rear end regulation cursor **4C** moves in the second direction **D2**, it does not contact the setting plate **1**. In other words, viewed from above, even if the sheet **S** is smaller than the setting plate **1**, the width direction regulation cursors **4A** and **4B** can contact the edges of the sheet **S** without contacting the setting plate **1**, and the rear end regulation cursor **4C** can contact the edge of the sheet **S** without contacting the setting plate **1**. In FIG. **6**, the setting plate **1** is shown with hatching.

(57) The width direction regulation cursor **4A** contacts the edge of the sheet **S** from the front side in the first direction **D1**, and the width direction regulation cursor **4B** contacts the edge of the sheet **S** from the back side in the first direction **D1**. Specifically, the width direction regulation cursors **4A** and **4B** each have a plate-like side surface part **40** that contacts the edge of the sheet **S**. The side surface part **40** of each of the width direction regulation cursors **4A** and **4B** stands in the up and down direction, and its thickness direction is the first direction **D1**. The side surface parts **40** of the width direction regulation cursors **4A** and **4B** face each other in the first direction **D1**.

(58) The width direction regulation cursors **4A** and **4B** move in the first direction **D1** in an interlocking manner. When the width direction regulation cursor **4A** moves backward, the width direction regulation cursor **4B** moves frontward, and hence an interval between the width direction regulation cursors **4A** and **4B** in the first direction **D1** becomes smaller. When width direction regulation cursor **4A** moves frontward, the width direction regulation cursor **4B** moves backward, and hence the interval between the width direction regulation cursors **4A** and **4B** in the first direction **D1** becomes larger.

(59) In order to move the width direction regulation cursors **4A** and **4B** in the first direction **D1** in an interlocking manner, as illustrated in FIG. **7**, the width direction regulation cursor **4A** is provided with a rack **401** that extends backward in the first direction **D1**, and the width direction regulation cursor **4B** is provided with a rack **402** that extends frontward in the first direction **D1**. Between the racks **401** and **402** in the second direction **D2**, there is disposed a gear **400** that rotates about an axis extending in the up and down direction. The gear **400** engages with the racks **401** and **402**.

(60) Note that when moving the width direction regulation cursors **4A** and **4B** in the first direction

D1, the user operates width direction regulation cursor **4A** so as to move the width direction regulation cursor **4A** in the first direction **D1**. In this structure, the width direction regulation cursor **4A** corresponds to a “regulation cursor”.

(61) <Structure of Hold Mechanism>

(62) As illustrated in FIGS. **8** to **12**, the sheet storage device **100** includes a hold mechanism **5** that stationarily holds the width direction regulation cursor **4A** at any position in the first direction **D1**. The hold mechanism **5** has a function of regulating positional deviation of the width direction regulation cursor **4A** in the first direction **D1**. When the positional deviation of the width direction regulation cursor **4A** is regulated, positional deviation of the width direction regulation cursor **4B** is also regulated. In other words, the hold mechanism **5** stationarily holds not only the width direction regulation cursor **4A** but also the width direction regulation cursor **4B** at any position.

(63) The width direction regulation cursor **4A** has a plate-like bottom surface part **41** that extends from a lower end of the side surface part **40** frontward in the first direction **D1**. By bending a single metal sheet, the width direction regulation cursor **4A** having the side surface part **40** and the bottom surface part **41** is formed. The bottom surface part **41** has a link opening **410**. The link opening **410** is a through hole that penetrates the bottom surface part **41** in the thickness direction (i.e., the thickness direction of a plate-like part forming the bottom surface part **41**).

(64) The link opening **410** has a link protrusion **411** on its inside. The link protrusion **411** protrudes from an inner edge of the link opening **410** frontward in the first direction **D1**. The link protrusion **411** has a plate-like shape.

(65) The hold mechanism **5** includes a guide shaft **51**. The guide shaft **51** is a round bar. The guide shaft **51** extends in the first direction **D1**. The guide shaft **51** is disposed below the bottom surface part **41**. One end part of the guide shaft **51** on the front side in the first direction **D1** is fixed to an edge part of a bottom frame **Fb**, and the other end part of the same on the back side in the first direction **D1** is fixed to an attaching part **50** provided to a bottom surface of the bottom frame **Fb** (see FIGS. **6** and **7**). The bottom frame **Fb** is made of metal sheet and is a frame that forms a bottom part of the sheet storage device **100**.

(66) Note that a trolley member **403** is attached to the width direction regulation cursor **4A**. The trolley member **403** includes a roller **403a**, and can travel on the bottom surface of the bottom frame **Fb** in the first direction **D1**, together with the width direction regulation cursor **4A**. For instance, the bottom surface of the bottom frame **Fb** (i.e., the upper surface) is provided with a guide rail **GL**. The guide rail **GL** guides the trolley member **403** to travel in the first direction **D1**.

(67) The hold mechanism **5** includes a stopper **52**. The stopper **52** is made of metal sheet. The stopper **52** is linked with the width direction regulation cursor **4A**.

(68) The stopper **52** has a guide hole **521**. The guide hole **521** is a through hole penetrating the stopper **52** in the thickness direction (i.e., the thickness direction of a plate-like part forming the stopper **52**). The guide hole **521** has a circular hole shape. The guide shaft **51** is fixed to the bottom frame **Fb** in a state inserted in the guide hole **521**. The stopper **52** can reciprocate along the guide shaft **51** in the first direction **D1** in the state where the guide shaft **51** is inserted in the guide hole **521**.

(69) In addition, the stopper **52** has an end part **5202** on the opposite side of a hole side end part **5201** having the guide hole **521**, and allows the end part **5202** to protrude from the link opening **410** above the bottom surface part **41**. The end part **5202** corresponds to an “operation side end part”. In the following description, the end part **5202** is referred to as the operation side end part **5202**.

(70) The stopper **52** has a link hole **522** between the hole side end part **5201** and the operation side end part **5202**. The link hole **522** is a through hole penetrating the stopper **52** in the thickness direction. The link protrusion **411** is inserted in the link hole **522** from the back side in the first direction **D1**. By allowing the operation side end part **5202** to protrude from the link opening **410** above the bottom surface part **41**, and by inserting the link protrusion **411** in the link hole **522** from

the back side in the first direction D1, the stopper 52 is linked with the width direction regulation cursor 4A. The stopper 52 can reciprocate in the first direction D1 together with the width direction regulation cursor 4A.

(71) In addition, the hold mechanism 5 includes a coil spring 53. The coil spring 53 corresponds to a “biasing member”. The coil spring 53 is a compression spring. The coil spring 53 is fitted over the guide shaft 51. For instance, a bracket 54 is attached to a lower surface of the bottom surface part 41. The bracket 54 is positioned on the front side of the stopper 52 in the first direction D1. In addition, the bracket 54 has a through hole (no numeral) in which the guide shaft 51 is inserted.

(72) The coil spring 53 is sandwiched between the stopper 52 and the bracket 54 in the first direction D1. In this way, the coil spring 53 generates a biasing force that biases the stopper 52 backward in the first direction D1. The coil spring 53 biases the hole side end part 5201 backward in the first direction D1.

(73) <Stationary Holding of Width Direction Regulation Cursor>

(74) The guide hole 521 has an opening diameter larger than the diameter of the guide shaft 51. In this way, the stopper 52 can incline in the first direction D1 on a fulcrum of the hole side end part 5201. Here, the stopper 52 is biased backward in the first direction D1 at the hole side end part 5201 by the coil spring 53.

(75) Therefore, as illustrated in FIG. 11, when a lever 61 described later is not operated, the stopper 52 inclines on a fulcrum of the hole side end part 5201, to the front side in the first direction D1 (corresponding to “one side in a predetermined direction”). Note that as described later in detail, by operating the lever 61, the stopper 52 can be raised backward in the first direction D1 from the state where the stopper 52 is inclined frontward in the first direction D1.

(76) By inclining the stopper 52 frontward in the first direction D1, an inner edge of the guide hole 521 contacts the guide shaft 51, and hence a frictional resistance occurs. Then, this frictional resistance restricts the stopper 52 from moving in the first direction D1. In addition, as the stopper 52 is linked with the width direction regulation cursor 4A, when the stopper 52 is restricted from moving in the first direction D1, the width direction regulation cursor 4A is restricted from moving in the first direction D1.

(77) In this embodiment, the width direction regulation cursor 4A is moved in the first direction D1 so that the side surface part 40 of the width direction regulation cursor 4A contacts the edge of the sheet S, and then the stopper 52 is inclined frontward in the first direction D1. In this way, the width direction regulation cursor 4A is stationarily held. In other words, the width direction regulation cursor 4A can be stationarily held at any position in the first direction D1. In this case, the frictional resistance, due to contact between the inner edge of the guide hole 521 and the guide shaft 51, can suppress positional deviation of the width direction regulation cursor 4A in the first direction D1.

(78) In addition, in this embodiment, as the coil spring 53 biases the hole side end part 5201 backward in the first direction D1, the stopper 52 can be prevented from rising backward in the first direction D1. In this way, in the state where the width direction regulation cursor 4A contacts the sheet S (i.e., in the state where positional deviation of the width direction regulation cursor 4A in the first direction D1 is restricted), it is possible to suppress positional deviation of the width direction regulation cursor 4A in the first direction D1, due to unintentional rising of the stopper 52 backward in the first direction D1.

(79) Here, in this embodiment, the biasing force of the coil spring 53 to bias the hole side end part 5201 to the back side in the first direction D1 is smaller than the frictional resistance generated when the inner edge of the guide hole 521 contacts the guide shaft 51. In this way, when the stopper 52 is inclined frontward in the first direction D1, it is possible to prevent the stopper 52 from being moved backward in the first direction D1 by the biasing force of the coil spring 53 while inclining frontward in the first direction D1. In other words, it is secured that the width direction regulation cursor 4A can be stationarily held.

(80) In addition, in this embodiment, by allowing the operation side end part **5202** to protrude from the link opening **410** above the bottom surface part **41**, and by inserting the link protrusion **411** in the link hole **522** from the back side in the first direction **D1**, even if the stopper **52** inclines frontward in the first direction **D1**, it is possible to prevent the linking between the width direction regulation cursor **4A** and the stopper **52** from being released. In other words, even if the stopper **52** inclines frontward in the first direction **D1**, the link protrusion **411** is not detached from the link hole **522**, and hence the linking between the width direction regulation cursor **4A** and the stopper **52** is not released.

(81) <Restriction of Inclination of Stopper>

(82) The stopper **52** has a contact part **523** in an integrated manner. The stopper **52** has the contact part **523** as its part. The contact part **523** of the stopper **52** protrudes from the link opening **410** and is disposed at a higher position than the bottom surface part **41**. The contact part **523** is a part that overhangs from the inner edge of the link opening **410** backward in the first direction **D1**. By bending a single metal sheet, the stopper **52** having the contact part **523** can be obtained.

(83) Here, when the stopper **52** is raised backward in the first direction **D1** from the state where it is inclined frontward in the first direction **D1**, so that the stopper **52** stands (or substantially stands), the contact part **523** contacts an upper surface of the bottom surface part **41**. In other words, the state illustrated in FIG. **11** changes to the state illustrated in FIG. **12**. When the stopper **52** is in the standing state, the frictional resistance due to contact between the inner edge of the guide hole **521** and the guide shaft **51** becomes smaller than that in the state where the stopper **52** is inclined frontward in the first direction **D1**, and hence the width direction regulation cursor **4A** can move with the stopper **52** in the first direction **D1**.

(84) In other words, in the state where the width direction regulation cursor **4A** can move in the first direction **D1**, the contact part **523** contacts the upper surface of the bottom surface part **41** (i.e., the width direction regulation cursor **4A**). By contacting the upper surface of the bottom surface part **41**, the contact part **523** restricts the stopper **52** from inclining backward in the first direction **D1**, when the width direction regulation cursor **4A** can move in the first direction **D1**.

(85) In this embodiment, by providing the stopper **52** with the contact part **523**, it is easy to keep the state where the width direction regulation cursor **4A** can move in the first direction **D1**, and it is possible to prevent unintentional restriction of movement of the width direction regulation cursor **4A** in the first direction **D1**. In other words, while enabling the width direction regulation cursor **4A** to be stationarily held at any position in the first direction **D1**, it is possible to prevent lowering of operability when moving the width direction regulation cursor **4A**. Note that this is convenient for the user because it is not necessary to be conscious about adjustment of force to be applied on the lever **61** described later.

(86) In addition, by providing the stopper **52** with the contact part **523**, it is not necessary to add another member for restricting inclination of the stopper **52** to the back side in the first direction **D1**. In this way, it is possible to prevent the structure from being complicated. In other words, cost increase can be suppressed.

(87) <Operation of Stopper>

(88) As illustrated in FIGS. **8**, **11**, and **12**, the sheet storage device **100** includes an operation mechanism **6**. The operation mechanism **6** is a mechanism for operating the stopper **52**. The operation mechanism **6** is attached to the width direction regulation cursor **4A**. The operation mechanism **6** is disposed outward of the width direction regulation cursor **4A** in the first direction **D1**. The operation mechanism **6** includes the lever **61** and a relay member **62**.

(89) The lever **61** is supported by a rotation shaft **610** in a rotatable manner. The rotation shaft **610** is attached to the width direction regulation cursor **4A** so as to extend in the second direction **D2**. In other words, the lever **61** is rotatable about an axis extending in the second direction **D2**. The lever **61** receives an operation by the user. By operating the lever **61**, the width direction regulation cursors **4A** and **4B** can be moved in the first direction **D1**.

(90) A position of an upper end part **611** of the lever **61** on an upper side of the rotation shaft **610** in the up and down direction is substantially the same as a position of an upper surface part of the width direction regulation cursor **4A** in the up and down direction. The upper surface part of the width direction regulation cursor **4A** has a recess (no numeral) that is recessed inward from the outside in the first direction **D1**. The upper end part **611** of the lever **61** is disposed inside the recess. In the operation of the lever **61**, the upper end part **611** is moved. For instance, the upper end part **611** of the lever **61** is provided with a plastic grip. A lower end part **612** of the lever **61** on a lower side with respect to the rotation shaft **610** is disposed at a position capable of contacting the relay member **62**.

(91) The relay member **62** transmits a user's operation of the lever **61** to the stopper **52**. The relay member **62** is supported by a rotation shaft **620** in a rotatable manner. The rotation shaft **620** is attached to the width direction regulation cursor **4A** below the rotation shaft **610**, so as to extend in the second direction **D2**. In other words, the relay member **62** is rotatable about an axis extending in the second direction **D2**. A position of the rotation shaft **620** in the first direction **D1** is the same as a position of the rotation shaft **610** in the first direction **D1**.

(92) The relay member **62** has an input end part **621** that is an upper side end part than the rotation shaft **620**, and an output end part **622** that is a lower side end part than the rotation shaft **620**. In other words, the relay member **62** has a support part **62a** between the input end part **621** and the output end part **622**. The support part **62a** of the relay member **62** is supported in a rotatable manner about an axis extending in the second direction **D2**.

(93) The lower end part **612** of the lever **61** contacts the input end part **621** from the back side in the first direction **D1**. The operation side end part **5202** of the stopper **52** contacts the output end part **622** from the back side in the first direction **D1**.

(94) When moving the width direction regulation cursor **4A** in the first direction **D1**, the user operates the lever **61**. In this case, the upper end part **611** of the lever **61** is moved backward in the first direction **D1**. When the upper end part **611** of the lever **61** moves backward in the first direction **D1**, the lower end part **612** of the lever **61** moves frontward in the first direction **D1**.

(95) Here, the lever **61** allows its lower end part **612** to contact the input end part **621** of the relay member **62** from the back side. Therefore, when the lower end part **612** of the lever **61** moves frontward in the first direction **D1**, the lower end part **612** of the lever **61** presses the input end part **621** frontward in the first direction **D1**. In other words, when the lever **61** is operated, it presses the input end part **621** frontward in the first direction **D1**. When the input end part **621** is pressed frontward in the first direction **D1**, the relay member **62** moves the output end part **622** backward in the first direction **D1**.

(96) The relay member **62** allows its output end part **622** to contact the operation side end part **5202** of the stopper **52** from the front side in the first direction **D1**. Therefore, when the output end part **622** moves backward in the first direction **D1**, the output end part **622** presses the operation side end part **5202** backward in the first direction **D1**. In other words, when the lever **61** presses the input end part **621** frontward in the first direction **D1**, the relay member **62** presses the operation side end part **5202** backward in the first direction **D1** by the output end part **622**.

(97) When the operation side end part **5202** is pressed backward in the first direction **D1**, the stopper **52** is raised backward in the first direction **D1** from the state where it is inclined frontward in the first direction **D1** (see FIG. 11). Finally, the stopper **52** becomes the standing state (see FIG. 12), and the contact part **523** of the stopper **52** contacts the upper surface of the bottom surface part **41**. In this way, the stopper **52** is not further inclined backward in the first direction **D1**. In other words, the width direction regulation cursor **4A** is maintained in the state where it can move in the first direction **D1**.

(98) In this embodiment, as the operation mechanism **6** can adjust inclination of the stopper **52**, it is easy to move the width direction regulation cursors **4A** and **4B** in the first direction **D1**, and it is possible to stationarily hold the width direction regulation cursors **4A** and **4B** at any position in the

first direction D1. It is only necessary for the user to move the upper end part 611 of the lever 61, and hence operability can be improved.

(99) <Stationarily Hold of Rear End Regulation Cursor>

(100) Although not illustrated, the rear end regulation cursor 4C is linked with the hold mechanism 5, which has the same structure as the hold mechanism 5 linked with the width direction regulation cursor 4A. In other words, the rear end regulation cursor 4C also corresponds to the “regulation cursor”. The width direction regulation cursor 4A corresponds to a “first regulation cursor”, and the rear end regulation cursor 4C corresponds to a “second regulation cursor”. In addition, focusing on the rear end regulation cursor 4C as the “regulation cursor”, the right side in the second direction D2 corresponds to “one side in a predetermined direction”, while the left side in the second direction D2 corresponds to “the other side in a predetermined direction”.

(101) In this structure, it is possible to regulate not only positional deviation of the width direction regulation cursors 4A and 4B in the first direction D1, but also positional deviation of the rear end regulation cursor 4C in the second direction D2. In other words, it is secured that the rear end regulation cursor 4C can be stationarily held at any position in the second direction D2. In addition, when the rear end regulation cursor 4C is moved in the second direction D2, movement of the rear end regulation cursor 4C in the second direction D2 is not unintentionally restricted. In this way, it is possible to suppress lowering of operability when moving the rear end regulation cursor 4C in the second direction D2.

(102) The embodiment disclosed in this specification is merely an example in every aspect, and should not be interpreted as a limitation. The scope of the present disclosure is defined not by the above description of the embodiment but by the claims, and should be understood to include all modifications within meaning and scope equivalent to the claims.

Claims

1. A sheet storage device comprising: a setting plate on which a sheet is set; a regulation cursor supported in a manner capable of reciprocating in a horizontal direction, the regulation cursor being configured to contact an edge of the sheet on the setting plate from a predetermined direction, so as to regulate positional deviation of the sheet in the predetermined direction; and a hold mechanism configured to stationarily hold the regulation cursor, wherein the hold mechanism includes a guide shaft extending in the predetermined direction, and a stopper linked with the regulation cursor, the stopper having a guide hole in which the guide shaft is inserted, and being capable of reciprocating with the regulation cursor along the guide shaft, the stopper is capable of inclining to one side in the predetermined direction on a fulcrum of a hole side end part having the guide hole, when the stopper is inclined to the one side, an inner edge of the guide hole contacts the guide shaft so as to generate frictional resistance, which restricts movement of the regulation cursor with the stopper, and when the stopper is raised to the other side opposite to the one side from the state where movement of the regulation cursor is restricted, the regulation cursor is enabled to move with the stopper, the stopper has a contact part, and the contact part contacts the regulation cursor, so as to restrict the stopper from inclining to the other side, when the regulation cursor is enabled to move.
2. The sheet storage device according to claim 1, further comprising an operation mechanism for operating the stopper, wherein the operation mechanism includes a lever configured to receive an operation, and a relay member configured to transmit the operation of the lever to the stopper, the relay member includes an input end part for the lever to contact from the other side, an output end part for an operation side end part of the stopper on the opposite side of the hole side end part to contact from the other side, and a support part between the input end part and the output end part, the support part being supported in a rotatable manner about an axis extending in a direction perpendicular to the predetermined direction, when the lever is operated, it presses the input end part to the one side, and when the input end part is pressed to the one side, the relay member

presses the operation side end part to the other side, by the output end part.

3. The sheet storage device according to claim 2, wherein the hold mechanism includes a biasing member, and the biasing member biases the hole side end part to the other side.

4. The sheet storage device according to claim 3, wherein a biasing force of the biasing member biasing the hole side end part to the other side is smaller than the frictional resistance.

5. The sheet storage device according to claim 2, wherein the regulation cursor includes a plate-like side surface part standing in an up and down direction, so as to contact the edge of the sheet on the setting plate from the predetermined direction, and a plate-like bottom surface part extending from a lower end of the side surface part to the one side, the guide shaft is disposed below the bottom surface part, the bottom surface part has a link opening penetrating the bottom surface part in the thickness direction, the link opening has a link protrusion protruding from an inner edge of the link opening to the one side, the stopper has a link hole between the hole side end part and the operation side end part, and by allowing the operation side end part to protrude from the link opening above the bottom surface part, and by inserting the link protrusion in the link hole from the other side, the stopper is linked with the regulation cursor.

6. The sheet storage device according to claim 5, wherein the stopper has the contact part at a part protruding from the link opening so as to be positioned above the bottom surface part, and the contact part overhangs from the inner edge of the link opening to the other side, and contacts an upper surface of the bottom surface part, so as to restrict the stopper from inclining to the other side.

7. The sheet storage device according to claim 1, wherein a first regulation cursor and a second regulation cursor are each linked to the hold mechanism, the first regulation cursor is supported in a movable manner in a first direction as the predetermined direction, so as to regulate positional deviation of the sheet in the first direction, and the second regulation cursor is supported in a movable manner in a second direction perpendicular to the first direction as the predetermined direction, so as to regulate positional deviation of the sheet in the second direction.

8. An image formation system comprising: the sheet storage device according to claim 1; and an image forming apparatus linked with the sheet storage device, wherein the sheet storage device feeds the image forming apparatus with the sheet, and the image forming apparatus performs printing on the sheet fed from the sheet storage device.
